

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – IV
PAPER –2
TEST DATE: 18-01-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

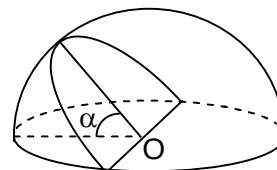
Physics

PART – I

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. The strength of the electric field produced by charges uniformly distributed over the surface of a hemisphere at its centre O is E_0 . A part of the surface is isolated from this hemisphere by two planes passing through the same diameter and forming an angle α with each other. Determine the electric field strength E produced at the same point O by the charges located on the isolated surface (on the “mericarp”).



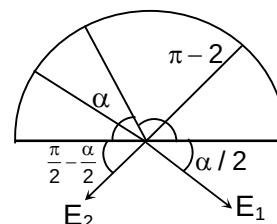
- (A) $\frac{E_0}{2} \sin \frac{\alpha}{2}$
- (B) $\frac{2E_0}{3} \sin \frac{\alpha}{2}$
- (C) $E_0 \sin \frac{\alpha}{2}$
- (D) $2E_0 \sin \frac{\alpha}{2}$

Ans. C

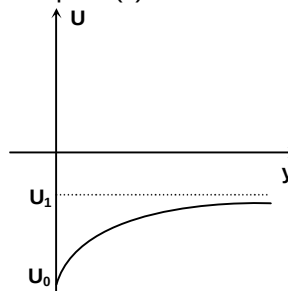
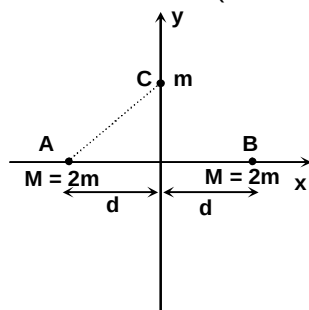
Sol. $E_1 \cos \frac{\alpha}{2} = E_2 \sin \frac{\alpha}{2}$ (i)

$E_1 \sin \frac{\alpha}{2} + E_2 \cos \frac{\alpha}{2} = E_0$ (ii)

Solving we get E_1



2. Two point masses, each of mass M are kept at rest at points A and B respectively. A third point m is released from infinity with a negligible speed, so that it can move along y-axis under the influence of mutual gravitational attraction on it due to point masses kept at A and B respectively as shown in the figure -1. Figure -2 represents the potential energy of system (includes m , M at A and M at B) with position of m at y-axis. (Neglect any other forces other than gravity) (given $Gm^2/d = 12 \text{ Joule}$ ($m = 6 \text{ kg}$). Choose the correct option(s)



- (A) Point mass m will perform periodic motion
- (B) $U_1 = -24$ Joule
- (C) Maximum speed of particle is 24 m/s
- (D) Maximum speed of particle is 4 m/s

Ans. A, B, D

Sol.
$$U = \frac{-2GMm}{\sqrt{d^2 + y^2}} + \frac{-GM^2}{2d}$$

$$= \frac{-2GMm}{\sqrt{d^2 + y^2}} - \frac{2Gm^2}{d}$$

$$= -2Gm^2 \left[\frac{2}{\sqrt{d^2 + y^2}} + \frac{1}{d} \right]$$

$$= |\Delta U| = \frac{1}{2}mv^2$$

3. A coil of radius R carries a current I . Another concentric coil of radius r ($r < R$) carries current $\frac{I}{2}$. Initially planes of the two coils are mutually perpendicular and both the coils are free to rotate about common diameter. They are released from rest from this position. The masses of the coils are M and m respectively ($m < M$). During the subsequent motion let K_1 and K_2 be the maximum kinetic energies of the two coils respectively and let U be the maximum magnitude of potential energy of magnetic interaction of the system of the coils. Choose the correct options.

- (A) $\frac{K_1}{K_2} = \frac{M}{m} \left(\frac{R}{r} \right)^2$
- (B) $K_1 = \frac{Umr^2}{mr^2 + MR^2} K_2 = \frac{UMR^2}{mr^2 + MR^2}$
- (C) $U = \frac{\mu_0 \pi I^2 r^2}{4R}$
- (D) $K_2 \gg K_1$

Ans. B, C, D

Sol. $\tau_1 = \tau_2$

$$\frac{\int \tau dt}{I} = \omega$$

$$\omega \propto \frac{1}{I}$$

$$\text{K.E.} = \frac{1}{2} I \omega^2 \propto \frac{1}{I}$$

$$\frac{K_1}{K_2} = \frac{\frac{mr^2}{2}}{\frac{MR^2}{2}}$$

$$\frac{K_1}{K_2} = \frac{m}{M} \left(\frac{r}{R} \right)^2$$

$$K_1 + K_2 = U$$

$$K_2 \frac{m}{M} \left(\frac{r}{R} \right)^2 + K_2 = U$$

$$K_2 = \frac{UMR^2}{mr^2 + MR^2}$$

$$K_1 = \frac{Umr^2}{mr^2 + MR^2}$$

$$U = \frac{1}{2} \pi r^2 \frac{\mu_0 I}{2R} = \frac{\mu_0 I^2 \pi r^2}{4R}$$

4. A uniform rod of mass m and length L performs small oscillations about horizontal axis passing through its upper end. Find the mean kinetic energy and angular amplitude β of rod, if at $t = 0$ it is deflected from vertical by an angle θ_0 and imparted an angular velocity ω_0 .

(A) $\beta = \sqrt{\frac{2L\omega_0^2}{3g} + \theta_0^2}$

(B) $\langle k_f \rangle = \frac{3}{4} MgL \sqrt{\frac{4L\omega_0^2}{3g} + \theta_0^2}$

(C) $\langle k_f \rangle = \frac{1}{8} MgL \sqrt{\frac{2L\omega_0^2}{3g} + \theta_0^2}$

(D) $\beta = \sqrt{\frac{3L\omega_0^2}{2g} + \theta_0^2}$

Ans. A, C

Sol. $\omega = \sqrt{\frac{Mg\ell}{I}}$

Where $I = \frac{ML^2}{3}$ and $\ell = \frac{L}{2}$

Thus $\omega = \sqrt{\frac{Mg(L/2)}{ML^2/3}} = \sqrt{\frac{3g}{2L}}$

$$\omega_0 = \sqrt{\frac{3g}{2L}} \sqrt{\beta^2 - \theta_0^2}$$

$$\beta = \sqrt{\frac{2L\omega_0^2}{3g} + \theta_0^2}$$

$$\begin{aligned}
 < k_f > = \frac{1}{4} l \omega^2 \beta^2 \\
 \text{or} \quad &= \frac{1}{4} \left(\frac{ML^2}{3} \right) \left(\frac{3g}{2L} \right) \sqrt{\frac{2L\omega_0^2}{3g} + \theta_0^2} \\
 \text{or} \quad &= \frac{1}{8} MgL \sqrt{\frac{2L\omega_0^2}{3g} + \theta_0^2}
 \end{aligned}$$

5. The resistive force suffered by a motor boat is $\propto V^2$ when the engine was shut down. When the velocity is V_0 find the average velocity at any time t .

(A) $V_{av} = \frac{V_0 + V}{2}$

(B) $\frac{VV_0}{2(V_0 + V)}$

(C) $\frac{VV_0 \log_e \frac{V_0}{V}}{(V_0 - V)}$

(D) $\frac{2VV_0 \log_e \frac{V_0}{V}}{(V_0 + V)}$

Ans. C

Sol. $\frac{dV}{dt} = -kV^2, \int_{V_0}^V \frac{dV}{V^2} = \int_0^t k dt \left[\frac{1}{V} - \frac{1}{V_0} \right] = kt$

or $t = \frac{1}{K} \left[\frac{1}{V} - \frac{1}{V_0} \right]$

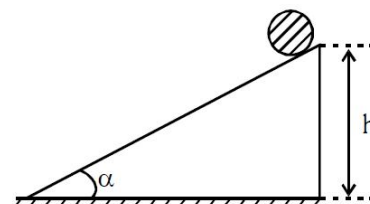
$\frac{dV}{dx} \cdot \frac{dx}{dt} = -kV^2$ or $\int_{V_0}^V \frac{dV}{V} = \int_0^x -k dx$

$\log_e V_0 - \log_e V = kx \quad \frac{\log_e V_0 / V}{k} = x$

or $V_{av} = \frac{x}{t} = \frac{(\log_e V_0 / V) \times k}{k \left[\frac{1}{V} - \frac{1}{V_0} \right]}$

or $V_{av} = \frac{VV_0 \log_e \frac{V_0}{V}}{(V_0 - V)}$

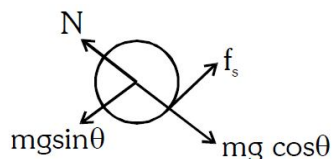
6. A disc of mass 'm' is released from top of an inclined plane of inclination α as shown in figure. If the coefficient of friction between inclined surface and disc is μ . Then choose the correct statement (s)



- (A) If $\mu > \frac{1}{3} \tan \alpha$, the net force acting on disc is $\frac{2mg \sin \alpha}{3}$
- (B) If $\mu > \frac{1}{3} \tan \alpha$, the velocity of disc at the bottom of inclined plane is $\sqrt{\frac{4gh}{3}}$
- (C) If disc rolls on inclined plane without slipping, then frictional force is $\frac{mg \sin \alpha}{3}$
- (D) If $\mu < \frac{1}{3} \tan \alpha$, the acceleration of disc along the inclined plane is $g \sin \alpha$

Ans. A, B, C

Sol. For pure rolling $mg \sin \alpha - f_s = ma$ and $f_s r = \frac{mr^2}{2} \alpha$, $a = \alpha r$



On solving $a = \frac{2g \sin \alpha}{3}$ and $f_s = \frac{mg \sin \alpha}{3}$

For no slipping, $\frac{mg \sin \alpha}{3} \leq \mu mg \cos \alpha$

And If $\mu < \frac{\tan \alpha}{3}$, slipping will be there so, $a = g \sin \alpha - \frac{f_k}{m} < g \sin \alpha$

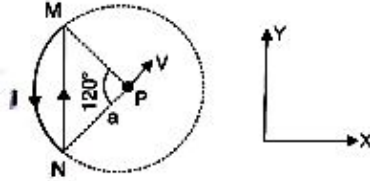
And velocity of disc $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ gives $v = \sqrt{\frac{4gh}{3}}$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

A wire loop carrying current I is placed in the X-Y plane as shown in the figure. If a particle with charge $+Q$ and mass m is placed at the centre P and given a velocity along NP (figure).



7. Find the instantaneous acceleration

- (A) $\frac{QV \mu_0 I}{m 6a} \left(\frac{3\sqrt{3}}{\pi} - 2 \right)$
- (B) $\frac{QV \mu_0 I}{m 4a} \left(\frac{3\sqrt{3}}{\pi} - 1 \right)$
- (C) $\frac{QV \mu_0 I}{m 6a} \left(\frac{3\sqrt{3}}{\pi} - 1 \right)$
- (D) None of these

Ans. C

Sol. $\vec{B}$ at centre $= (\vec{B}_{\text{st. wire}} - \vec{B}_{\text{arc}})(-\hat{k})$

$$= \left[\frac{\mu_0}{4\pi R \cos 60} (2 \sin 60) - \frac{\mu_0}{4\pi R} \left(\frac{2\pi}{3} \right) \right] (-\hat{k})$$

$$= \frac{\mu_0 I}{6a} \left(\frac{3\sqrt{3}}{\pi} - 1 \right) (-\hat{k})$$

For charge particle $-qvB = ma$

$$\Rightarrow a = \frac{qv\mu_0 I}{6ma} \left(\frac{3\sqrt{3}}{\pi} - 1 \right)$$

8. If an external uniform magnetic field $\vec{B} = B\hat{i}$ is applied, find the torque acting on the loop due to the field.

- (A) $\vec{\tau} = BI \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right) a^2 \hat{j}$
- (B) $\vec{\tau} = BI \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right) a^2 \hat{j}$

$$(C) \quad \vec{\tau} = BI \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right) a^2 \hat{j}$$

(D) None of these

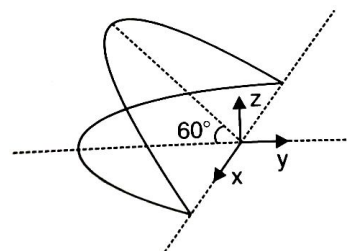
Ans. A

$$\text{Sol.} \quad \vec{T} = \vec{M} \times \vec{B} = NIA(\hat{k}) \times B\hat{i}$$

$$\Rightarrow \vec{T} = I \left(\frac{\pi}{3} a^2 - \frac{\sqrt{3}}{4} a^2 \right) B \hat{j}$$

Paragraph for Question Nos. 09 and 10

A circular loop of wire of radius 'a' is bent about its diameter such that the angle between the two halves becomes 60° . The resistance per unit length of the wire is r_0 . A uniform magnetic field directed along (+) Z-axis, is varying as $B = B_0 t$. (Neglect self inductance of the loops)



9. Magnitude of induced emf in the loop is

$$(A) \quad \frac{B_0 \pi a^2}{8}$$

$$(B) \quad \frac{B_0 \pi a^2}{4}$$

$$(C) \quad \frac{B_0 \pi a^2}{2}$$

$$(D) \quad 2B_0 \pi a^2$$

Ans. B

Sol. For loop PQR,

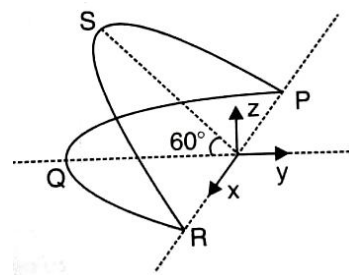
$$\phi_1 = \vec{B} \cdot \vec{A} = B_0 t A \cos 0^\circ = B_0 t \left(\frac{\pi a^2}{2} \right)$$

$$|e_1| = \left| \frac{d\phi_1}{dt} \right| = \frac{B_0 \pi a^2}{2}$$

$$\text{For loop PSR, } \phi_2 = B_0 t \left(\frac{\pi a^2}{2} \right) \cos 60^\circ$$

$$\text{So } |e_2| = \left| \frac{d\phi_2}{dt} \right| = \frac{B_0 \pi a^2}{4}$$

$$|e| = |e_1| - |e_2| = \frac{B_0 \pi a^2}{4}$$



10. If the system is maintained in the position shown at all the time the external torque need to keep the loop in the position shown at any time is

(A) $\frac{\sqrt{3}B_0^2\pi a^3 t}{32r_0}$

(B) $\frac{\sqrt{3}B_0^2\pi^2 a^4 t}{16r_0}$

(C) $\frac{B_0^2\pi^2 a^4 t}{16r_0}$

(D) $\frac{B_0^2\pi a^3 t}{32r_0}$

Ans. A

Sol. $i = \frac{e}{2\pi a r_0} = \frac{B_0 a}{8r_0}$

For loop PQR, $\tau = 0$

For loop PSR, $\tau = MB \sin 60^\circ = (iA)(B_0 t) \left(\frac{\sqrt{3}}{2} \right) = \left(\frac{B_0 a}{8r_0} \right) \left(\frac{\pi a^2}{2} \right) (B_0 t) \left(\frac{\sqrt{3}}{2} \right)$

$= \frac{\sqrt{3}B_0^2\pi a^3 t}{32r_0}$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

11. The prism spectrum is spread out more at violet end than at the red end. Angular dispersion is defined as $\phi = \frac{d\delta}{d\lambda}$, if deviation of small prism δ is defined as $\delta = (\mu - 1)A_0$ and wavelength is defined as $\mu = A + \frac{B}{\lambda^2}$, where A and B are constants and A_0 is prism angle, then dispersion $\phi \propto \frac{1}{\lambda^N}$. Find the value of N.

Ans. 3

Sol. As, $\phi = \frac{d\delta}{d\lambda}$

This can be written as

$\phi = \frac{d\delta}{d\mu} \times \frac{d\mu}{d\lambda}, \delta = (\mu - 1)A_0$

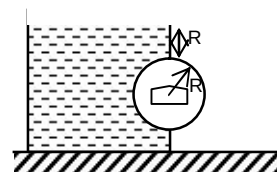
$\frac{d\delta}{d\mu} = A_0$

$$\mu = A + \frac{B}{\lambda^2} \Rightarrow \frac{d\mu}{d\lambda} = -\frac{2B}{\lambda^3}$$

$$\text{So, } \phi = -\frac{2BA_0}{\lambda^3} \Rightarrow \phi \propto \frac{1}{\lambda^3}$$

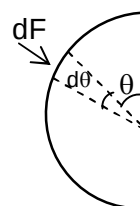
$$\text{So, } N = 3$$

12. A cylinder of radius R is kept embedded along the wall of a dam as shown. Take density of water as ρ and length of cylinder as L . The vertical force exerted by water on the cylinder is $\frac{\rho g L R^2 \pi}{X}$. Find the value of X .



Ans. 2

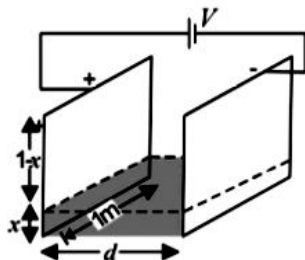
$$\begin{aligned} \text{Sol. } f_v &= \int_0^R \rho g (2R - R \cos \theta) (L R d\theta) \cos \theta \\ &= L \rho g R^2 \int_0^R (2 \cos \theta - \cos^2 \theta) d\theta \\ &= \rho g L R^2 \frac{\pi}{2} \end{aligned}$$



13. Two square metal plates of side 1 m are kept 0.01 m apart like a parallel plate capacitor in air in such a way that one of their edges is perpendicular to an oil surface in a tank filled with an insulating oil. The plates are connected to a battery of emf 564 V. The plates are then lowered vertically into the oil at a speed of 0.001 ms^{-1} . Calculate the current (in nano ampere) drawn from the battery during the process. (Dielectric constant of oil = 11, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-1}$)

Ans. 5

Sol. The adjacent figure is a case of parallel plate capacitor. The combined capacitance will be



$$\begin{aligned} C &= C_1 + C_2 \\ &= \frac{k\epsilon_0(x \times 1)}{d} + \frac{\epsilon_0[(1-x) \times 1]}{d} \end{aligned}$$

$$C = \frac{\epsilon_0}{d} [dx + 1 - x] \quad \dots(i)$$

After time dt , the dielectric rises by dx . The new equivalent capacitance will be

$$\begin{aligned} C + dC &= C_1' + C_2' \\ &= \frac{\epsilon_0}{d} [(x + dx) \times 1] + \frac{\epsilon_0[(1 - x - dx) \times 1]}{d} \end{aligned}$$

$$= \frac{\epsilon_0}{d} [kx + kdx + 1 - x - dx] \quad \dots(ii)$$

We know that $q = CV$

$$\frac{dq}{dt} = V \frac{dC}{dt} \quad \dots(iii)$$

$$\Rightarrow I = V \frac{\epsilon_0}{d} (k-1)v$$

From (i) and (ii)

$$I = \frac{564 \times 8.85 \times 10^{12}}{0.01} (11-1) \times 0.001 \cong 5 \times 10^{-9} \text{ A}$$

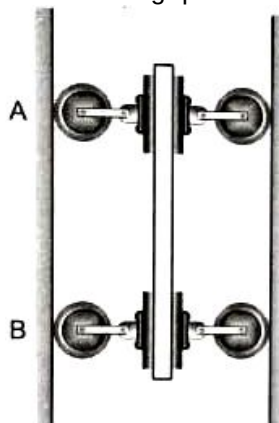
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

A bar of mass m is held as shown between 4 uniform discs, each of mass m' and radius $r = 75 \text{ mm}$. The bar has been released from rest, knowing that the normal forces exerted on the discs by vertical wall are sufficient to prevent any slipping. Answer the following questions. (Acceleration due to gravity is 10 m/s^2)



14. If $m = 5 \text{ kg}$ and $m' = 2 \text{ kg}$, what is acceleration of the bar

Ans. 7.65

Sol. F.B.D. of disc

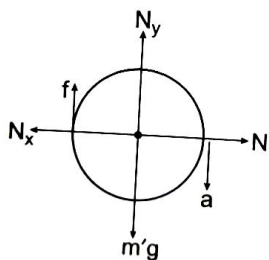
$$\Rightarrow \text{In horizontal direction } N = N_x \quad \dots(i)$$

$$\text{In vertical direction } m'g - f - N_y = m'a \quad \dots(ii)$$

Torque due to friction;

$$fR = \frac{m'R^2}{2} \alpha$$

$$\text{As } a = R\alpha$$



$$\therefore fR = \frac{m'R^2}{2} \frac{a}{R}$$

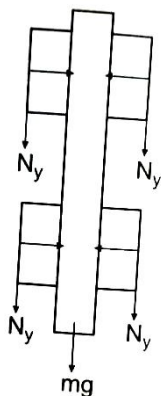
$$\Rightarrow fR = \frac{m'R^2}{2} \frac{a}{R}$$

$$\Rightarrow f = \frac{m'a}{2} \text{ put in (ii)}$$

$$\therefore m'a - N_y = \frac{3m'a}{2}$$

$$N_y = m'g - \frac{3m'a}{2} \quad \dots(\text{iii})$$

Now, F.B.D. of bar;



$$4N_y + mg = ma$$

Put N_y from (iii)

$$4m'a - 6m'a + mg = ma \quad \dots(\text{iv})$$

As $m' = 2 \text{ kg}$ and $m = 5 \text{ kg}$

$$\therefore 8g - 12a + 5g = 5a$$

$$a = \frac{13g}{17} \downarrow = \frac{130}{17}$$

15. If the mass m of the bar is negligible, then acceleration of the bar is

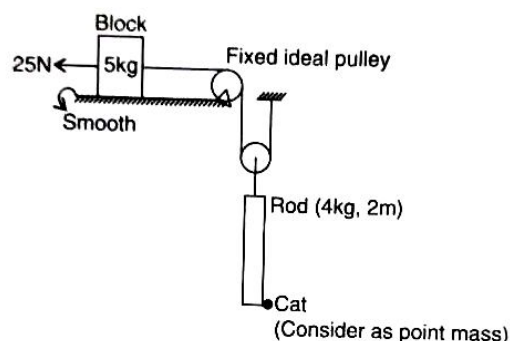
Ans. 6.67

Sol. From equation (iv) (neglecting m) $a = \frac{2g}{3} \downarrow = \frac{20}{3}$

Question Stem for Question Nos. 16 and 17

Question Stem

A cat initially hanging on a rod in static equilibrium as shown in figure. If the cat climbs up on the rod from its lower end towards its upper end with a monotonically increasing acceleration starting at $t = 0$ and crossing the midpoint of rod at $t = t_0$ and the average speed of cat in this time interval ($t = 0$ to $t = t_0$) is 2m/sec relative to the rod then in the time interval $t = 0$ to $t = t_0$:
(Use SI units)



16. If the angle made by displacement of center of mass of (block + cat) is ϕ with horizontal, then $\tan \phi$ is

Ans. 2.40

17. The distance (in meter) travelled by the block is

Ans. 0.08

Sol. (for Q. 16 & 17)

As system was at equilibrium initially

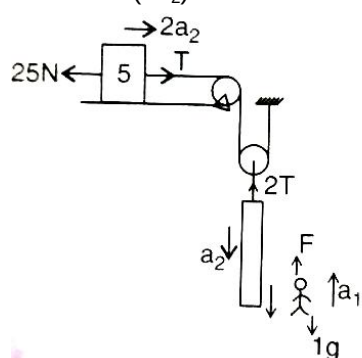
$$\text{So } m_{\text{cat}} = 1 \text{ kg}$$

$$\text{Now } f - 1g = |a|$$

(where f is the force on cat applied by rod)

$$4g + f - 2T = 4a_2$$

$$T - 25 = 5(2a_2)$$



Solving these question:

$$a_1 = 24 a_2$$

So, average velocity will have same relation

$$v_1 = 24v_2$$

Relative velocity of cat w.r.t. rod

$$2 \text{ m/s} = v_1 + v_2$$

$$\text{So, } v_2 = (2 / 25) \text{ m/sec}$$

$$v_1 = (48/25) \text{ m/sec}$$

$$\text{For displacement of cat } t_0 = \frac{1}{v_1 + v_2} = \frac{1}{2} \text{ s}$$

$$\tan \phi = \frac{M_{\text{cat}} Y_{\text{cat}}}{M_{\text{block}} X_{\text{block}}} = \frac{\left(\frac{48}{25}\right) \times 1}{\left(\frac{4}{25}\right) \times 5} = \frac{12}{5} \quad \dots(i)$$

$$\text{Displacement of block} = \frac{4}{25} \times \frac{1}{2} = \frac{2}{25}$$

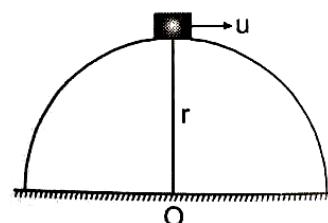
Question Stem for Question Nos. 18 and 19

Question Stem

A small block of mass m is projected horizontally from the top of the smooth and fixed hemisphere of radius $r = 8 \text{ m}$ with speed u as shown. For values of $u \geq u_0$, ($u_0 = \sqrt{gr}$) it does not slide on the hemisphere.

[i.e., leaves the surface at the top itself]

18. For $u = 2 u_0$, it lands at point P on ground. Find OP:



Ans. 22.63

Sol. $mg = \frac{mu_0^2}{r} \Rightarrow u_0 = \sqrt{gr}$

Now, along vertical; $r = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2r}{g}}$

Along horizontal; $OP = 2u_0 t = 2\sqrt{2}r$

19. For $u = u_0/3$, find the height from the ground at which it leaves the hemisphere:

Ans. 5.63

Sol. As at B it leaves the hemisphere,

$\therefore N = 0$

$$mg \frac{h}{r} = \frac{mv^2}{r}$$

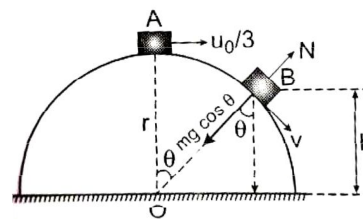
$$mv^2 = mgh$$

... (i)

By energy conservation between A and B

$$mgr + \frac{1}{2}m\left(\frac{u_0}{3}\right)^2 = mgh + \frac{1}{2}mv^2$$

Put u_0 and $mv^2 \therefore h = \frac{19r}{27}$



Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

20. By reacting with which of the following reagents CH_3NH_2 will form a solid derivative?

- (A) PhSO_2Cl
- (B) $(\text{COOEt})_2$
- (C) H_2SO_4
- (D) $\text{CHCl}_3 + \text{KOH}$

Ans. A, B, C

Sol. With (A) solid sulphonamide, with (B) solid substituted oxamide with (C) solid salt.

21. Which of the following is /are soluble in alkaline water?

- (A) Me_3CCOOH
- (B) $(\text{COOH})_2$
- (C) PhCOOH
- (D) HCOOH

Ans. A, B, C, D

Sol. Because of salt formation all are soluble.

22. Which of the following will form phthalic anhydride by following this scheme?

(P) $\text{KMnO}_4, \text{OH}^-$

(Q) H_3O^+

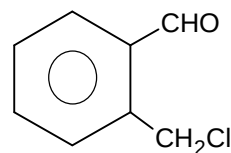
(R) Δ

(A) o-xylene

(B) m-xylene

(C) p-xylene

(D)



Ans. A, D

Sol. Only (A) and (D) on oxidation form phthalic acid.

23. Nylons is/are characterized by:

- (A) high melting solid polymer
- (B) high degree of crystallinity
- (C) hydrophobic
- (D) high mechanical

Ans. A, B, D

Sol. Can form H-bond with H_2O and hence are hydrophilic.

24. Which of the following statement(s) is/are true?

- (A) In metal carbonyl complexes d_{C-O} increases compared to that in CO molecule.
- (B) The pair of compounds $[Cr(H_2O)_6Cl_3]$ and $[CrCl_3(H_2O)_3].3H_2O$ show hydrate isomerism.
- (C) d_{z^2} orbital of central metal atom/ion is used in dsp^2 hybridization.
- (D) Facial and Meridional isomers associated with $[Ma_3b_3]^{n\pm}$ type complex compound, both are optically inactive.

Ans. A, D

Sol. Factual

25. Correct statement is/are:

- (A) an acidified solution of $K_2Cr_2O_7$ liberates iodine from KI
- (B) $K_2Cr_2O_7$ is used as a standard solution for estimation of Fe^{2+} ions
- (C) in acidic medium $N = M \times 6$ for $K_2Cr_2O_7$.
- (D) $(NH_4)_2Cr_2O_7$ on heating decomposes to yield Cr_2O_3 through an endothermic reaction

Ans. A, B, C, D

Sol. $Cr_2O_7^{2-} + I^- \rightarrow Cr^{3+} + I_2$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 26 and 27

Green compound as mineral (A) decomposes on heating to give black residue (B) and gaseous compound (C) and (D). When (D) come in contact of CuSO_4 , it gives blue colour. (C) gives milkiness with lime water and it is colourless and odourless gas. (B) reacts with sulphuric acid to give soluble compound which can react with potassium ferrocyanide to give red brown ppt. (E).

26. Compound (E) is:

- (A) $\text{Cu}_2[\text{Fe}(\text{CN})_6]$
- (B) $\text{Zn}_2[\text{Fe}(\text{CN})_6]$
- (C) $\text{Cl}_2[\text{Fe}(\text{CN})_6]$
- (D) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$

Ans. A

27. Compound (B) + $\text{H}_2\text{SO}_4 \rightarrow (\text{X}) \xrightarrow[\text{excess}]{\text{NH}_4\text{OH}} (\text{Y})$, compound (Y) is:

- (A) $[\text{Zn}(\text{NH}_3)_4]\text{SO}_4$
- (B) $\text{Cu}(\text{OH})_2$
- (C) $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
- (D) $[\text{Cd}(\text{NH}_3)_4]\text{SO}_4$

Ans. C

Sol. (for Q. 26-27)

- (A) Malachite
- (B) CuO
- (C) CO_2
- (D) H_2O
- (E) $\text{Cu}_2[\text{Fe}(\text{CN})_6]$

Paragraph for Question Nos. 28 and 29

Covalent radius is half of the distance between two nuclei of two like atoms bonded together by a single covalent bond. Considering the homonuclear diatomic molecule A_2 , covalent radius, $r_A = \frac{d_{A-A}}{2}$, where d_{A-A} is bond length but if covalent bond is formed between two elements of different electronegativity, then we use the following relation, given by Stevenson in 1941. $d_{A-B} = r_A + r_B - 0.09(X_A - X_B)$ X_A and X_B are electronegativities of A and B. (E. N. of Cl = 3.0)

28. If bond length of Cl_2 is $1.98\text{\AA}$, then covalent radius of Cl atoms is:

- (A) $0.99\text{\AA}$
- (B) $<0.99\text{\AA}$
- (C) $>0.99\text{\AA}$
- (D) None of these

Ans. A

29. If bond length A_2 molecule is $0.8\text{\AA}$ and ΔEN of A-Cl is 1.0 then calculate bond length of A-Cl:

- (A) $1.39\text{\AA}$
- (B) $1.30\text{\AA}$
- (C) $2.78\text{\AA}$
- (D) $2.60\text{\AA}$

Ans. B

Sol. (for Q. 28-29)

$$r_A = 0.4\text{\AA}$$

$$r_{Cl} = 0.99\text{\AA}$$

$$d_{A-B} = r_A + r_B - 0.09(X_A - X_B)$$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

30. For a certain reaction the variation of the rate constant with temperature is given by the equation

$$\ln K_t = \ln K_0 + \left(\frac{\ln 3}{10}\right)T \quad (T \geq 0^\circ\text{C})$$

The value of the temperature coefficient of the reaction rate is therefore.

Ans. 3

Sol. $\ln\left(\frac{K_t}{K_0}\right) = \frac{t}{10}(\ln 3)$: when $t = 10$;

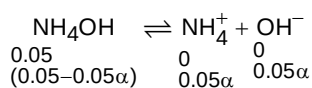
$$\frac{K_t}{K_0} = \text{temperature coefficient.}$$

$$\therefore \ln(\text{temperature coefficient}) = (\ln 3) \times \frac{10}{10} = \ln 3$$

31. Determine percentage degree of dissociation of 0.05 M NH_3 at 25°C in a solution of $\text{pH} = 11$.

Ans. 2

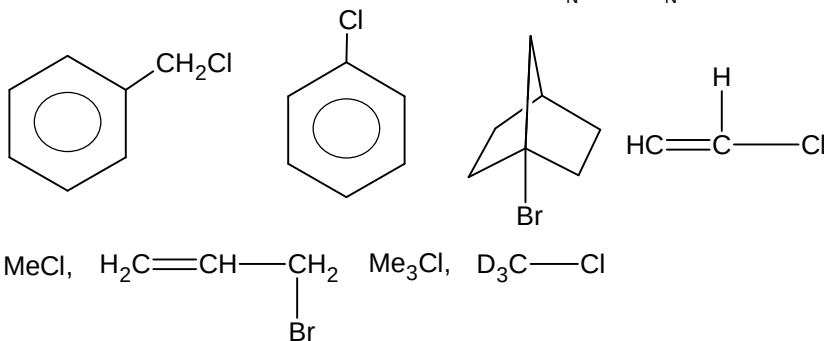
Sol.



$$\text{pH} = 11, [\text{H}^+] = 10^{-11}$$

$$[\text{OH}^-] = 10^{-3} = C\alpha$$

32. The number of substrates which neither show $\text{S}_{\text{N}}1$ nor $\text{S}_{\text{N}}2$.



Ans. 3

Sol. 2, 3, and 4

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

In an ionic compound A^+X^- , radii of A^+ and X^- ions are 1.0 pm and 2.0 pm, respectively. The volume of the unit cell of the crystal AX is $x \text{ pm}^3$.

33. Find the value of $\frac{x}{5}$

Ans. 43.20

34. Coordination number of cations is

Ans. 6.00

Sol. (for Q.33 & 34)

$$\frac{r^+}{r^-} = \frac{1}{2} = 0.5 \quad [\text{FCC}] \rightarrow \text{cations in octahedral void.}$$

$$r^+ + r^- = \frac{a}{2}$$

$$a = 6$$

$$\text{Volume} = a^3$$

Question Stem for Question Nos. 35 and 36

Question Stem

If the slope of 'Z' (compressibility factor) vs. 'P' curve is constant $\left(\text{slope} = \frac{\pi}{492.6} \text{ atm}^{-1}\right)$ at a particular temperature (300 K) and very high pressure,

35. Calculate radius of the molecules. (in Å)

Ans. 2.50

36. Find the value of vander waal constant "b"

Ans. 0.16

Sol. $Z = 1 + \frac{Pb}{RT}$ at high pressure

$$\text{Slope} = \frac{b}{RT} = \frac{\pi}{492.6}$$

$$\Rightarrow b = \frac{\pi}{492.6} \times 0.0821 \times 300 \text{ and}$$

$$b = \frac{4}{3} \pi r^3 \times 4N_A$$

Question Stem for Question Nos. 37 and 38

Question Stem

0.10 g of a sample containing CuCO_3 and some inert impurity was dissolved in dilute sulphuric acid and volume made up to 50 mL. This solution was added into 50 mL of 0.04M KI solution where copper precipitates as CuI and I^- is oxidized into I_3^- . A 10 mL portion of this solution is taken for analysis, filtered and made up free I_3^- and then treated with excess of acidic permanganate solution. Liberated iodine required 20 mL of 2.5 mM sodium thiosulphate solution to reach the end point.

37. Determine mass percentage of CuCO_3 in the original sample.

Ans. 74.10

38. Find milimoles of I^- which are reacted with CuCO_3

Ans. 1.50

Sol. M-moles of KI = $50 \times 0.04 = 2$
m-equivalent of $\text{Na}_2\text{S}_2\text{O}_3$ reacted with 10 mL solution = $20 \times 2.5 \times 10^{-3} = 50 \times 10^{-3} = 50 \times 10^{-3}$
m-equivalent of I^- left unreacted in 100 mL solution = $50 \times 10^{-3} \times \frac{100}{10} = 0.5$
m-moles of I^- which reacts with $\text{CuCO}_3 = 2 - 0.5 = 1.5$
 $\therefore 2\text{Cu}^{2+} + 5\text{I}^- \rightarrow 2\text{CuI} + \text{I}_3^-$
m-moles of $\text{Cu}^{2+} = \frac{2}{5} \times 1.5 = 0.6$
weight of CuCO_3 $0.6 \times 10^{-3} \times 123.5 = 0.0741$
weight % of $\text{CuCO}_3 = \frac{0.0741}{0.1} \times 100$
 $= 74.1$

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. Let z_1, z_2, z_3 be the coordinate of vertices A, B, C of the triangle if $W_1 = z_1 - z_2$, $W_2 = z_3 - z_2$ and $\angle A = 90^\circ$, then

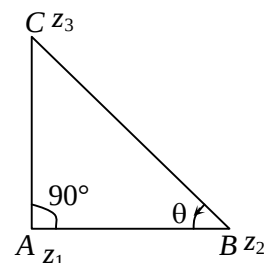
- (A) $\operatorname{Re}\left(\frac{W_2}{W_1}\right) = 1$
- (B) $\operatorname{Re}(W_1 \bar{W}_2) = |W_1|^2$
- (C) $\operatorname{Re}(W_1 \bar{W}_2) = |W_2|^2$
- (D) $\operatorname{Re}\left(\frac{W_1}{W_2}\right) = 1$

Ans. A, B

Sol. $\frac{W_2}{|W_2|} = \frac{W_1}{|W_1|} e^{-i\theta}$

$$\Rightarrow \frac{W_2}{|W_2|} = \frac{|W_2|}{|W_1|} e^{-i\theta} = 1 - i \tan \theta$$

$$W_1 \bar{W}_2 = \frac{W_1}{W_2} |W_2|^2 = |W_1| |W_2| e^{i\theta} = (|W_1|^2 + i |W_1| |W_2| \sin \theta)$$



40. Let $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n}$, where n is odd integer, if

$$S_1 = a_0 + a_4 + a_8 + \dots$$

$$S_2 = a_1 + a_5 + a_9 + \dots$$

$$S_3 = a_2 + a_6 + a_{10} + \dots$$

$$S_4 = a_3 + a_7 + a_{11} + \dots \text{ then}$$

- (A) $S_1 = S_3$
- (B) $S_2 = S_4$
- (C) $S_2 + S_4 = 0$
- (D) either $S_1 = S_2 = S_3$ or $S_1 = S_3 = S_4$

Ans. A, D

Sol. $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n} \quad \dots(1)$

$$\Rightarrow a_0 + a_2 + a_4 \dots = \frac{3^n + 1}{2}$$

$$a_1 + a_3 + a_5 \dots = \frac{3^n - 1}{2}$$

Put $x = i$ in equation (1) where $i = \sqrt{-1}$

If $n = 4m + 1$

$$\Rightarrow a_1 - a_3 + a_5 \dots = 1$$

$$a_0 - a_2 + a_4 \dots = 0$$

So, $a_1 + a_5 + a_9 \dots = \frac{3^n + 1}{4} = S_2$

$$a_3 + a_7 + a_{11} \dots = \frac{3^n - 3}{4} = S_4$$

Similarly if $n = 4m + 3$

Then, $S_2 = \frac{3^n - 3}{4}$ and $S_4 = \frac{3^n + 1}{4}$

41. Let $f_1(x) = \sin^{-1}(\cos(\sin^2 x))$, $f_2(x) = \cos^{-1}(\sin(\cos^2 x))$
 $f_3(x) = \sin^{-1}(\cos(\cos^2 x))$, $f_4(x) = \cos^{-1}(\sin(\sin^2 x))$, then choose the correct option(s)

(A) $\sum_{i=1}^4 f_i(x) = 2\pi - 2$

(B) $\sum_{i=1}^4 (-1)^i f_i(x) = 0$

(C) $\sum_{i=1}^4 (-1)^{\left[\frac{i}{3}\right]} f_i(x) = 0$ (where $[.]$ is greater integer function)

(D) $f_1(x) > f_2(x) \Rightarrow \cos 2x > 0$

Ans. A, B, C, D

Sol. $f_1(x) = \frac{\pi}{2} - \sin^2 x$, $f_2(x) = \frac{\pi}{2} - \cos^2 x$

$$f_3(x) = \frac{\pi}{2} - \cos^2 x, \quad f_4(x) = \frac{\pi}{2} - \sin^2 x$$

42. If the area of a triangle is given Δ and angle C is given and if the value of the side C opposite to angle C is minimum then

(A) $a = \sqrt{\frac{2\Delta}{\sin C}}$

(B) $b = \sqrt{\frac{2\Delta}{\sin C}}$

(C) $a = \frac{4\Delta}{\sin C}$

(D) $b = \frac{4\Delta}{\sin C}$

Ans. A, B

Sol. $c^2 = a^2 + b^2 - 2ab \cos C$
 $= (a - b)^2 + 2ab(1 - \cos C)$

$$\Delta = \frac{1}{2} ab \sin C$$

$$\text{or } 2ab = \frac{4\Delta}{\sin C}$$

$$\text{here } c^2 = (a - b)^2 + 4\Delta \left(\frac{1 - \cos C}{\sin C} \right)$$

$$= (a - b)^2 + 4\Delta \tan \frac{C}{2}$$

$$4\Delta \tan \frac{C}{2} \text{ is constant so for } c^2 \text{ to be minimum } a = b \text{ thus } 2ab = 2a^2 = \frac{4\Delta}{\sin C}$$

$$\Rightarrow a = b = \sqrt{\frac{2\Delta}{\sin C}}$$

43. $f : [1, 4] \rightarrow [7, 14]$ is a surjective, twice differentiable function such that $f''(x) > 0$,
 $\forall x \in (1, 4)$, then the equation $(f'(x))^2 = \frac{49}{9}$ has

(A) at least one root in $(1, 4)$

(B) exactly one root in $(1, 4)$

(C) at least two roots in $(1, 4)$

(D) at most two roots in $(1, 4)$

Ans. A, D

Sol. Since $f''(x) > 0$ graph of $f(x)$ is concave upward.

Now $f'(x) = \pm \frac{7}{3}$ having a solution would mean line having slope $\pm \frac{7}{3}$ touching the curve.

The graph will strictly lie in the rectangle formed by $x = 1$, $x = 4$, $y = 7$, $y = 14$.

If f is monotonic the line parallel to either of diagonal of rectangle will touch the graph exactly once and if f is non-monotonic, then lines parallel to diagonals will touch the graph exactly once.

44. If $\frac{2}{1!13!} + \frac{2}{3!11!} + \frac{2}{5!9!} + \frac{1}{7!7!} = \frac{2^m}{n!}$ then

- (A) $m + n = 27$
- (B) $m^2 = 1 + 2n$
- (C) $n^2 - m^2 = 27$
- (D) $n = 1 + m$

Ans. A, C, D

Sol. $\frac{2}{1!13!} + \frac{2}{3!11!} + \frac{2}{5!9!} + \frac{1}{7!7!} = \frac{1}{14!}(2^{14}C_1 + 2^{14}C_3 + 2^{14}C_5 + 2^{14}C_7) = \frac{1}{14!}2^{14-1} = \frac{2^{13}}{14!} \Rightarrow m = 13, n = 14$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively of two distinct positive numbers. For $n \geq 2$, let A_{n-1} and H_{n-1} has arithmetic, geometric and harmonic means as A_n, G_n, H_n respectively.

45. Which of the following statements is a correct statement?

- (A) $G_1 > G_2 < G_3 > \dots$
- (B) $G_1 < G_2 < G_3 < \dots$
- (C) $G_1 = G_2 = G_3 = \dots$
- (D) $G_1 < G_3 < G_5 < \dots$ and $G_2 > G_4 > G_6 > \dots$

Ans. C

46. Which of the following statements is correct?

- (A) $A_1 > A_2 > \dots$
- (B) $A_1 < A_2 < A_3 < \dots$

$$(C) \quad A_1 > A_3 > A_5 > \dots \text{ and } A_2 < A_4 < A_6 < \dots$$

$$(D) \quad A_1 < A_3 < A_5 < \dots \text{ and } A_2 > A_4 > A_6 < \dots$$

Ans. A

Sol. (for Q. 45 and 46)

Let two distinct positive numbers be a and b.

$$A_1 = \frac{1}{2}(a+b), G_1 = \sqrt{ab}, H_1 = \frac{2ab}{a+b}$$

$$A_n = \frac{1}{2}(A_{n-1} + H_{n-1}), G_n = \sqrt{A_{n-1} H_{n-1}}$$

$$H_n = \frac{2A_{n-1}H_{n-1}}{A_{n-1} + H_{n-1}}$$

For $n \geq 2$.

$$A_n H_n = A_{n-1} H_{n-1}$$

$$\text{Thus } A_1 H_1 = A_2 H_2 = A_3 H_3 = \dots$$

$$\Rightarrow G_1^2 = G_2^2 = G_3^2 = \dots$$

$$G_1 = G_2 = G_3 = \dots$$

As a and b are distinct, $A_1 > H_1$

$$\Rightarrow A_1 > A_2 > H_2 > H_1$$

$$\Rightarrow A_1 > A_2 > A_3 > H_3 > H_2 > H_1$$

And so on.

$$\text{Thus, } A_1 > A_2 > A_3 > \dots$$

$$\text{and } H_1 < H_2 < H_3 < \dots$$

Paragraph for Question Nos. 47 and 48

Let V_r denotes the sum of first r terms of an arithmetic progression (A.P.) whose first term is r and the common difference is $(2r-1)$. Let $T_r = V_{r+1} - V_r - 2$;

$$Q_r = T_{r+1} - T_r, W_r = \frac{1}{T_r} + \frac{1}{4(r+1)} \text{ and } X_r = 3^{Q_r} \text{ for } r = 1, 2, \dots$$

47. The sum $V_1 + V_2 + \dots + V_n$ is:

$$(A) \quad \frac{1}{12} n(n+1)(3n^2 - n + 1)$$

$$(B) \quad \frac{1}{12} n(n+1)(3n^2 + n + 2)$$

$$(C) \quad \frac{1}{2} n(2n^2 - n + 1)$$

$$(D) \quad \frac{1}{3} (2n^3 - 2n + 3)$$

Ans. B

$$\begin{aligned}
 \text{Sol. } V_r &= \frac{r}{2} [2(r) + (r-1)(2r-1)] \\
 &= \frac{1}{2} (2r^3 - r^2 + r) \\
 \therefore \sum_{k=1}^n V_k &= \sum_{k=1}^n k^3 - \frac{1}{2} \sum_{k=1}^n k^2 + \frac{1}{2} \sum_{k=1}^n k \\
 &= \frac{1}{4} n^2 (n+1)^2 - \frac{1}{12} n(n+1)(2n+1) + \frac{1}{4} n(n+1) \\
 &= \frac{1}{12} n(n+1) [3n(n+1) - (2n+1) + 3] \\
 &= \frac{1}{12} n(n+1) (3n^2 + n + 2)
 \end{aligned}$$

48. T_r is always:

- (A) an odd number
- (B) an even number
- (C) a prime number
- (D) a composite number

Ans. D

$$\begin{aligned}
 \text{Sol. } T_r &= V_{r+1} - V_r - 2 \\
 &= [(r+1)^3 - r^3] - \frac{1}{2} [(r+1)^2 - r^2] + \frac{1}{2} [(r+1) - r] - 2 \\
 &= (3r^2 + 3r + 1) - \frac{1}{2} (2r + 1) + \frac{1}{2} - 2 \\
 &= 3r^2 + 2r - 1 = (3r - 1)(r + 1) \\
 \therefore T_r &\text{ is always a composite number.}
 \end{aligned}$$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. If $(\lambda, \lambda + 1)$ is an interior point of $\triangle ABC$ where $A = (0, 3)$, $B = (-2, 0)$ and $C = (6, 1)$, then the number of integral values of λ is/are

Ans. 2

Sol. Since the point lies inside the triangle, $\lambda - 8(\lambda + 1) + 2 < 0$, $\lambda + 3(\lambda + 1) - 9 < 0$ and $3\lambda - 2(\lambda + 1) + 6 > 0$. Solving these, we get $-\frac{6}{7} < \lambda < \frac{3}{2}$.

50. Angle between the tangents drawn from (1, 4) to the parabola $y^2 = 4x$ is $\frac{\pi}{m}$, where m is equal to

Ans. 3

Sol. Equation of the pair of tangents is $(y^2 - 4x)(16 - 4) = [4y - 2(x + 1)]^2$

$$\Rightarrow 12(y^2 - 4x) = 4(2y - x - 1)^2$$

$$\Rightarrow 3(y^2 - 4x) = 4y^2 + x^2 + 1 - 4xy + 2x - 4y$$

$$\Rightarrow x^2 + y^2 - 4xy + 14x - 4y + 1 = 0$$

If θ is the angle between these lines

$$\text{Then } \tan \theta = \frac{2\sqrt{(2)^2 - (1)(1)}}{(1) + (1)} = \sqrt{3}$$

$$\Rightarrow \theta = \frac{\pi}{3} = \frac{\pi}{m} \Rightarrow m = 3$$

51. If the lengths of the semi major axis of an ellipse is 68 and the eccentricity is $\frac{1}{2}$ then the area of the rectangle formed by joining the vertices of the latus rectum of the ellipse is equal to A, where A – 6930 is

Ans. 6

Sol. Let the equation of the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a^2 = (68)^2$ and $b^2 = (68)^2 \left(1 - \frac{1}{4}\right)$ vertices of the lateral recta are $\left(\pm ae, \pm \frac{b^2}{a}\right)$

$$\begin{aligned} \text{Area of the rectangle formed by joining these points} &= 2ae \times \frac{2b^2}{a} = 4b^2e = 4 \times 68 \times \frac{3}{4} \times \frac{1}{2} \\ &= 6 \end{aligned}$$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53

Question Stem

There are five different boxes and seven different balls. All the seven balls are to be distributed in the five boxes placed in a row so that any box can receive any number of balls.

52. If λ be the number of ways these balls can be distributed so that no box is empty and order of putting the balls in the boxes is considered then the value of $\frac{\lambda}{11}$ is

Ans. 6872.73

Sol. Select 4 gaps out of 6 gaps between the balls to create 5 distinct partitions. Then permute the 7 balls

$$\Rightarrow \text{total} = {}^6C_4 \times 7! = 15 \times 7$$

53. Suppose, all the balls are identical, then number of ways these balls can be distributed into these boxes if empty boxes are also allowed is k then the value of $\frac{3k}{4}$ is

Ans. 247.50

Sol. No. of non-negative integral solution of the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 7$

$$\Rightarrow \text{total is } {}^{11}C_7.$$

Question Stem for Question Nos. 54 and 55

Question Stem

By taking $B(-3, 0)$ and $C(3, 0)$ triangles PBC of perimeter 16 are constructed. If S is the locus of incentre of triangle ABC, then

54. Equation of S is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then b^2 is

Ans. 2.25

Sol. As $ID \perp BC$

$$\text{Now } x = BD - BO = 5 - b \Rightarrow b = 5 - x$$

$$\therefore C = 10 - b = 5 + x$$

$$\text{Now } \Delta = \sqrt{8(8-6)(8-5+x)(8-5-x)}$$

$$(8y)^2 = 16(3+x)(3-x)$$

$$\Rightarrow x^2 + 4y^2 = 9$$

55. A ray emanating from $\left(-\frac{3\sqrt{3}}{2}, 0\right)$ is incident on the curve S at P whose ordinate is $\frac{3}{2}$. The

equation of the reflected ray is $\frac{x}{a} + \frac{y}{b} = 1$ then b is

Ans. 1.50

Sol. Equation of s is $x^2 + 4y^2 = 9$

$$P \equiv \left(0, \frac{3}{2}\right)$$

$$\text{Foci are } \left(\pm \frac{3\sqrt{3}}{2}, 0\right)$$

$$\text{Equation of the reflected ray is } 2x + 2\sqrt{3}y = 3\sqrt{3}$$

Question Stem for Question Nos. 56 and 57

Question Stem

For a recursive relation in 'n' we may generate nth term free from any other term i.e. purely in 'n'. Based on this answer the following questions

56. If $(a_{n+1})^2 a_n + 2(a_{n+1}) - a_n = 0$ and $a_0 = 1$, then $\lim_{n \rightarrow \infty} 2^{n+4} (a_n)$ is

Ans. 12.56

Sol. Given equation converts to $a_n = \frac{2a_{n+1}}{1 - (a_{n+1})^2}$

$$\Rightarrow a_n = \tan\left(\frac{\pi}{2^{n+2}}\right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n 2^n = \frac{\pi}{4}$$

57. If $a_n = 2a_{n+1}^2 - 1$ and $a_0 = 0$, then $\lim_{n \rightarrow \infty} 2e^{a_n}$ (use $e = 2.72$)

Ans. 5.44

Sol. Given equation converts to $a_{n+1} = \sqrt{\frac{1+a_n}{2}}$

$$\Rightarrow a_n = \cos\left(\frac{\pi}{2^{n+1}}\right)$$

$$\text{Hence, } \lim_{n \rightarrow \infty} e^{\cos\left(\frac{\pi}{2^{n+1}}\right)} = e$$